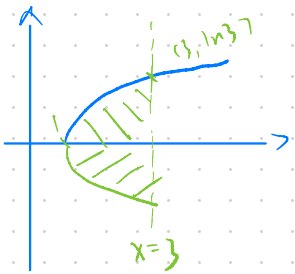
5–10 Set up, but do not evaluate, an integral for the volume of the solid obtained by rotating the region bounded by the given curves about the specified line.

5. $y = \ln x,\ y = 0,\ x = 3;$ about the x-axis

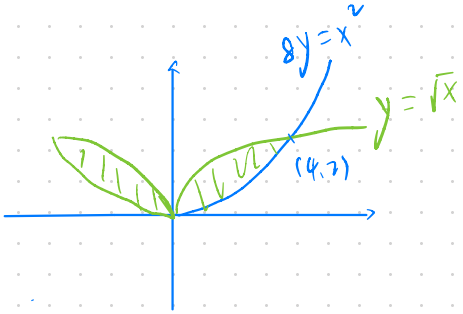


$$A = \int_1^3 \pi r^2\, dx$$
$$= \int_1^3 \pi \cdot (\ln x)^2\, dx$$

$y^2 = x$

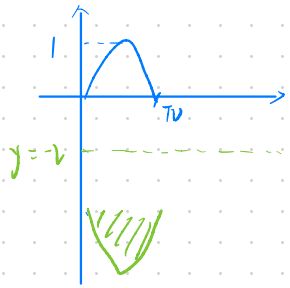
7. $8y = x^2,\ y = \sqrt{x};$ about the y-axis

$x = 8y$ $x = y^2$



$$A = \int_0^2 \pi r_{out}^2 - \pi r_{in}^2\, dy$$
$$= \int_0^2 \pi \cdot (2\sqrt{2}y)^2 - \pi \cdot (y^2)^2\, dy$$

9. $y = \sin x,\ y = 0,\ 0 \le x \le \pi;$ about $y = -2$

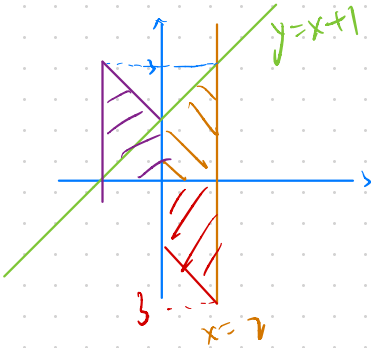


$$A = \int_0^\pi \pi r_{out}^2 - \pi r_{in}^2\, dx$$
$$= \int_0^\pi \pi \cdot (\sin x + 2)^2 - \pi \cdot 2^2\, dx$$

Exercises 6.2 5, 7, 9, 11, 13, 15, 17, 47, 49, 53, 55, 63.

11–28 Find the volume of the solid obtained by rotating the region bounded by the given curves about the specified line. Sketch the region, the solid, and a typical disk or washer.

11 $y = x + 1,\ y = 0,\ x = 0,\ x = 2;$ about the x-axis



$A = \int_0^2 2\pi r h\, dx$

$= \int_0^2 2\pi \cdot x \cdot (x+1)\, dx$ $\int x^2 + x$

$= \int_0^2 2\pi x^2 + 2\pi x\, dx$

$= 2\pi \cdot \frac{1}{3}[x^3]_0^2 + 2\pi \cdot \frac{1}{2}[x^2]_0^2$

$= \frac{2\pi}{3} \times 8 + \pi \times 4$

$= \frac{28}{3}\pi = 29.32\pi$

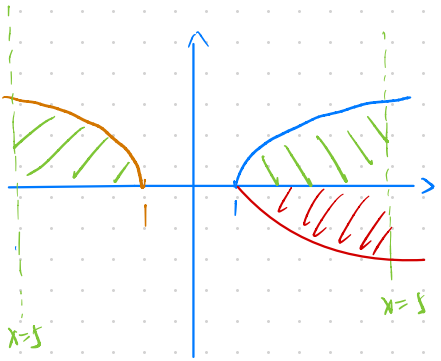
$A = \int_0^2 \pi r^2\, dx$

$= \int_0^2 \pi (x+1)^2\, dx$

$= \pi \int_0^2 x^2 + 2x + 1\, dx = \pi\left[\frac{x^3}{3} + x^2 + x\right]_0^2$

$= \frac{26}{3}\pi$

13. $y = \sqrt{x-1},\ y = 0,\ x = 5;$ about the x-axis



LIATE

$A = \int_1^5 2\pi r h \, dx$

$= \int_1^5 2\pi x \cdot \sqrt{x-1} \, dx$

$= 2\pi \int_1^5 x\sqrt{x-1} \, dx$

$\sqrt{x-1}$ $\frac{du}{dx} =$

$u = \sqrt{x-1}$

$dv = x\,dx$

$v = \frac{x^2}{2}$

Let $u = x-1$, $\frac{du}{dx} = 1 \Rightarrow du = dx$

when $x = 5$, $u = 4$

$x = 1$, $u = 0$

$= 2\pi \int_0^4$

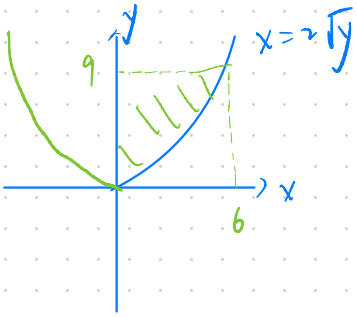
$A = \int_1^5 \pi r^2 \, dx = \int_1^5 \pi (\sqrt{x-1})^2 \, dx$

$= \pi \int_1^5 x-1 \, dx$

$= \pi \left[\frac{x^2}{2} - x \right]_1^5$

$= 8\pi$

15. $x = 2\sqrt{y},\ x = 0,\ y = 9;$ about the y-axis



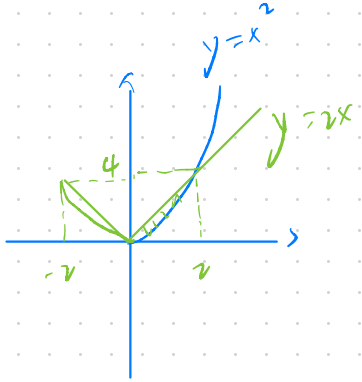
$$A = \int_0^9 \pi r^2\, dy$$

$$= \int_0^9 \pi \cdot (2\sqrt{y})^2\, dy$$

$$= 4\pi \int_0^9 y\, dy$$

$$= 4\pi \times \frac{1}{2}\left[y^2\right]_0^9 = 2\pi \times 81 = 162\pi$$

17. $y = x^2,\ y = 2x;$ about the y-axis



$$A = \int_0^2 2x - x^2\, dx$$

$$= \left[x^2 - \frac{x^3}{3}\right]_0^2$$

这样只有水平的一半

用 washer method

$$A = \int_0^4 \pi r_{out}^2 - \pi r_{in}^2\, dy$$

$$= \int_0^4 \pi(\sqrt{y})^2 - \pi\left(\frac{y}{2}\right)^2 dy$$

$$= \int_0^4 \pi y - \frac{\pi}{4} y^2\, dy$$

$$= \pi \int_0^4 y - \frac{1}{4} y^2\, dy = \pi\left[\frac{y^2}{2} - \frac{1}{12} y^3\right]_0^4 = \frac{8}{3}\pi$$

$r = x$

$r_{in} = x = \frac{y}{2}$

$r_{out} = x = \sqrt{y}$